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Grants as principal recipient

- PhD: Excellence funding for science/health research phd for 3 years “[Thèses Inria-Inserm Santé numérique](#)”, 61 200€ gross amount over 3 years.
- Scholarship on social criteria from 2008 to 2015: 36 800€ net amount over 8 years.
- Scholarship on excellence criteria from 2008 to 2015: 16 000€ net amount over 8 years.

University curriculum

- 2020 - today: Post doctoral student at Inria, Astral Team. Funding provided by Sartorius.
 - Developments of supervised missing data imputation.
 - Development of an outlier detection methodology.
 - Exchanges with research department of Sartorius.
- 2019: Phd in Biostatistics, under co-direction of J. Saracco and R. Thiébaud, Université de Bordeaux, France.
 - Analysis of Ebola vaccine (rVSV) research data-set.
 - First developments of a new PLS approach.
 - Applications to health and also agronomic problematic.
- 2015:
 - Engineer CentraleSupélec, MATIS («*Applied Mathematics for Information and Signal Processing*»), Gif-sur-Yvette, France.
 - Research master, Université Paris-Saclay, ATSI («*Automation and Signal and Image Processing*»), Orsay, France.

Collaborations

- [Corporate Research Advanced Data Analytics](#), Sartorius, Bordeaux: [Olivier Cloarec](#).
- [Statify team](#), Inria Grenoble Rhône-Alpes, Grenoble, France: [Stéphane Girard](#).
- [Astral team](#), Inria Bordeaux Sud-Ouest, Bordeaux: [Jérôme Saracco](#).
- [Biomarkers team](#), INRA, Bordeaux: [Marie-Pierre Ellies](#).
- [Vintage team](#), INSERM U1219, Bordeaux: [Misbah Razzaq](#), [Felipe Leal Valentim](#) and [David-Alexandre Trégouët](#).

- [SISTM team](#), INSERM U1219, Inria BSO, Bordeaux: [Rodolphe Thiébaud](#), [Laura Richert](#), [Solenne Delahaye](#), [Henri Bonnabau](#) and [Boris Hejblum](#).
- [University Medical Center Hamburg-Eppendorf](#), 1st Department of Medicine, 20246, Hambourg, Allemagne: [Anne Rechten](#).
- [International AIDS Vaccine Initiative](#), Londres, Royaume-Uni: Jonathan Hare.

Published articles

- GIRARD, S., LORENZO, H., SARACCO, J. «Advanced topics in Sliced Inverse Regression.», *Journal of Multivariate Analysis*, 2022, [📄](#)
- LORENZO, H., CLOAREC, O., RODOLPHE, T., SARACCO, J. «Data-Driven Sparse Partial Least Squares.», *Statistical Analysis and Data Mining*, 2022, [📄](#)
- ELLIES-OURY, M.-P. ,GRUFFAT, D., LISTRAT, A., LORENZO, H., CHAVENT, M., SARACCO, J. «The relationships between Sensory and Nutritional Quality are not consistent from one muscle to another in the same bovine carcass.», *Negative Results*, 2021, [📄](#)
- LORENZO, H., R. MISBAH, J. ODEBER, P.-E. MORANGE, J. SARACCO, D.-A. TRÉGOÛET AND R. THIÉBAUT. 2019, «High-dimensional multi-block analysis of factors associated with thrombin generation potential», *2019 IEEE 32nd International Symposium on Computer-Based Medical Systems (CBMS)*, IEEE, p. 453–458, [📄](#)
- ELLIES-OURY, M.-P.* , H. LORENZO* , C. DENOYELLE, J. SARACCO AND B. PICARD. 2019, «An original methodology for the selection of biomarkers of tenderness in five different muscles», *Foods*, vol. 8, n°6, p. 206, [📄](#)
- RECHTIEN, A.* , L. RICHERT*, H. LORENZO, ... , R. THIÉBAUT, M. ALTFLED AND M. ADDO. 2017, «Systems Vaccinology Identifies an Early Innate Immune Signature as a Correlate of Antibody Responses to the Ebola Vaccine rVSV-ZEBOV», *Cell reports*, vol. 20, n°9, p. 2251–2261., [📄](#)

Book chapter

- LORENZO, H., SARACCO, J. «Computational Outlier Detection Methods in Sliced Inverse Regression.», *Advances in Contemporary Statistics and Econometrics, Festschrift in Honor of Christine Thomas-Agnan*, Springer, p. 101-122, 2021, [📄](#)

Oral communications

- LORENZO, H., O. CLOAREC AND J. SARACCO. 2022, «Koh-Lanta, missing data imputation in supervised context», *Chimiométrie Brest*.

*. Same contribution from the authors.

- GIRARD, S., LORENZO, H. AND J. SARACCO. 2022, «Advanced topics in Sliced Inverse Regression», *50th Anniversary Jubilee Edition of JMVA, 2022*.
- LORENZO, H. AND J. SARACCO. 2022, «Detection of atypic observations in SIR regression», *Data-quitaine 2022*, Bordeaux.
- LORENZO, H., J. SARACCO AND C. WEINREICH. 2022, «Sélection de variables en régression SIR par seuillage doux ou seuillage dur de la matrice d'intérêt», *Journées de Statistique 2022 - Société Française de Statistique (SFdS)*, Lyon, France.
- LORENZO, H., O. CLOAREC AND J. SARACCO. 2021, «How to deal with missing values in the high dimensional supervised context?», *CMStatistics 2021, King's College*, London, UK. 
- LORENZO, H., R. THIÉBAUT, J. SARACCO AND O. CLOAREC. 2021, «Multiblock supervised analyses. Should we really normalize blocks?», *17th Scandinavian Symposium on Chemometrics - SSC17*, Aalborg, Denmark. 
- LORENZO, H., O. CLOAREC, R. THIÉBAUT AND J. SARACCO. 2021, «Détection d'individus atypiques en régression SIR», *Journées de Statistique 2021 - Société Française de Statistique (SFdS)*, Nice, France.
- LORENZO, H., O. CLOAREC, R. THIÉBAUT AND J. SARACCO. 2021, «A sparse PLS between Statistics and Machine Learning», *Journées de Statistique 2021 - Société Française de Statistique (SFdS)*, Nice, France.
- LORENZO, H., J. SARACCO AND R. THIÉBAUT. 2019, «High-Dimensional Multi-Block Analysis of Factors Associated with Thrombin Generation Potential», *32nd IEEE CBMS International Symposium on Computer-Based Medical Systems – CBMS2019*, Cordoue, Espagne. 
- LORENZO, H., R. THIÉBAUT AND J. SARACCO. 2018, «Multi-block high-dimensional lasso penalized analysis with imputation of missing data applied to postgenomic data in an Ebola vaccine trial», *Annual workshop on Statistical Methods for Post Genomic Data - SMPGD 2018*, Montpellier, France. 
- LORENZO, H., J. SARACCO AND R. THIÉBAUT. 2018, «Une PLS multivoie parcimonieuse avec observations manquantes pour données hétérogènes», *Journées de Statistique 2018 - Société Française de Statistique (SFdS)*, EDF Lab Paris Saclay, France. 

Poster communications

- LORENZO, H., O. CLOAREC AND J. SARACCO. 2022, «Imputation for supervised learning problems in high dimension», *Compstat 2022*, Bologne. 
- LORENZO, H., J. SARACCO ET R. THIÉBAUT. 2019, «Apprentissage supervisé pour données massives multi-blocs incomplètes», *JED 2019 - EDSP2*, Bordeaux. 

- HARE, J., C. STREATFIELD, J. YATES, W. KILEMBE, S. LAKHI, H. LORENZO, ... ET J. GILMOUR. 2016. «CD4 T-cell Counts More Closely Associate with the Cytokine Profiles Observed in Acutely Infected Volunteers than Viral Load or Replicative Capacity», *AIDS RESEARCH AND HUMAN RETROVIRUSES* (Vol. 32, pp. 170-170). 140 HUGUENOT STREET, 3RD FL, NEW ROCHELLE, NY 10801 USA: MARY ANN LIEBERT, INC. 

Seminars

- Inria, BootSir, 2022, Bordeaux. Organizing board, 10 participants.
- Corporate Research Advanced Data Analytics (Sartorius), How Koh-Lanta works, November 2021, Bordeaux, 12 participants.
- ENS of Lyon, L3, «Introduction to statistical learning», 21st of January 2020, Sept Laux, France, 30 participants.
- Sanofi Pasteur, a summary of my works, 21st of January 2020, Lyon, France, 15 participants.
- Biostatistics seminars, BPH U1219, Bordeaux, 30 participants. 
- SISTM team seminars, BPH U1219, Bordeaux, 15 participants.
- ISPED doctoral students' seminars, BPH U1219, Bordeaux, 20 participants.

Teaching

- [DU BDSI \(Big data and statistics for the engineer\)](#), Missing data treatment, ENSC, Bordeaux. Courses for adult students, class of 10 students.
 - 2021: Lecture and tutorial: 8 hours.
 - 2022: Lecture and tutorial, PLS and missing-data treatment: 2,5 full days.
- [L3 engineer - ENSEIRB-MATMECA](#), Probabilities and Statistics, Bordeaux. Classes of 30 students.
 - 2020-2021: 46 hours.
 - 2021-2022: 22 hours.
- [DU BDSI \(Big data and statistics for the engineer\)](#), ENSC, Bordeaux. Class of 8 students.
 - 2018-2019: Lecture and tutorial: 11 hours.
- [M2 engineer - Biostatistics](#), ENSAI, Rennes (Bruz), France. Class of 30 students.
 - 2018-2019: Genetic project: 21 hours.
- [M2 - Biostatistics](#), ISPED, Bordeaux. Class of 30 students.
 - 2017-2018: Lecture and tutorial: 26 hours.
 - 2018-2019: Lecture and tutorial: 6 hours.

- [Health and Science mixed-master](#), M1, ISPED, Bordeaux. Courses for L2 students, classes of 30 students, the objective was to quickly form medical students to “Mathematics” (Analysis, Algebra, Probabilities, Statistics and Coding).
 - 2018-2019: Lecture and tutorial: 24 hours.
- [ISPED Summer School](#), Bordeaux. Class of 30 students.
 - 2017-2018: Lecture and tutorial: 6 hours.

Supervision experience

- Clément Weinreich, **M1 student**, 3 months project, 2022-ongoing, development of the **SIR-thresholded** approach, participation to the JDS 2022 and collaboration to a paper.
- Arthur Saracco, **L3 student**, 3 months project, 2022, App. development.
- Alexandre Conanec, **PhD student**, steering committee of thesis, title «*Modelling of the optimisation of the management of beef production qualities and performances*», co-supervision by MP Ellies-Oury and J Sarracco.
- Arthur Saracco, **L3 student**, 1 months internship, June to July 2021. «*Création de Quizz d'évaluation de notions de probabilités et de statistique.*» Co-Supervision with J. Saracco.
- Valentin Kochegarov, **L2 student**, 3 months internship, May to July 2019. «*Etude des méthodes de l'analyse de l'expression différentielle des données de séquençage d'ARN à haut débit.*» Co-Supervision with R. Thiébaud.

Software productions

- Package **SIRthresholded** with an available [vignette](#) and packages present
 - on CRAN via <https://cran.r-project.org>
 - on GitHub via [Clement-W/SIR-thresholded](#).
- Package **ddsPLS**:
 - New R version (**C++** deep optimization code) on GitHub via <https://github.com/hlorenzo/ddsPLS2>.
 - Implementation of multiple imputation Koh-Lanta algorithm.
 - R version stable via [CRAN](#) and the associated [vignette](#).
 - Python version in development on GitHub via https://github.com/hlorenzo/py_ddspls.
- Package **outlierSIR** on GitHub via <https://github.com/hlorenzo/outlierSIR>
- Personal website (R blogdown) via <https://hadrienlorenzo.netlify.com/>.

Misc training & qualifications

- **Languages**: French native, English advanced, Czech intermediate, Italian intermediate.